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Structural, optical and piezoelectric investigation on new Brucinium Chlorate di-hydrate NLO single crystal for optoelectronic, piezo-sensor, transducer and OLED applications

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ABSTRACT

A new and optically good transparent Brucinium Chlorate di-hydrate (BCDH) single crystal was grown from aqueous solution by the slow evaporation technique at a pH of 5 with dimension $11 \times 4 \times 2\,mm^3$ over a period of 40–45 days. It belongs to the monoclinic crystal system with non-centrosymmetric space group C2 which is identified by single crystal XRD analysis. Percentage of carbon, hydrogen and oxygen are confirmed by chemical CHN analysis. The development faces were identified by morphological studies. The BCDH crystal possesses 78% transmittance which is confirmed by UV–Visible analysis. The presences of various functional groups were confirmed by vibrational analysis. Piezoelectric charge co-efficient (d_{33}) of BCDH crystal was found to be 4 p C/N. The frequency doubling ability of title compound was analyzed using Kurtz and Perry powder method and its output were compared with standard materials. The LDT value of the BCDH crystal was found to be 3.51 GW/cm^2. The electron excitation and life time measurements of the molecules in grown BCDH crystal was studied by Fluorescence spectral analysis. The stability and various decomposition stages of BCDH crystal were scrutinized by different thermal analysis.

1. Introduction

In recent scenario, the synthesis of new NLO material is a predominant chore of modern materials science for the growth of several single crystal based novel devices in the upcoming field of optoelectronics such as optical modulation, optical switching, optical logic, electro optic shutters and storage devices [1–6] for the developing technologies in telecommunications and signal processing. Especially in the case of electronic devices such as light emitting diodes (LEDs), field effect transistors (FETs) and photocells [7] these kinds of optical materials are necessary for its better performance. The rationalization, estimation and prediction of NLO behavior of the materials are thus a considerable interest in the various research fields. The second order non linear optical properties (i.e. second harmonic generation (SHG)) of the materials are very crucial for above mentioned applications. They are largely depend on many parameters such as inter and intra molecular interactions, dipole moments [7], hyperpolarizability and symmetric nature of the sample etc. In order to increase the second-order NLO property of the macroscopic level, first one should enhance the NLO performance at the molecular level of the compound [8].

The structure of alkaloid Brucine is unique to strychnine [9], which is a natural host mix with six asymmetric carbon atoms, does not contain H-bonding donor group and it has a tendency to crystallize with the acid group containing guest molecules [10]. $CuCl_2$ is a paramagnetic and most of its derivatives are due to the Jahn-Teller effect exhibit the distortion from idealized octahedral geometry property. For the past decades, its complexes with amino acids have attracted a lot of attentions [11,12].

The intention of the present communication is that, the BCDH crystal was grown and different investigation such as Structural, Chemical, Morphological, Optical, Vibrational, LDT Piezoelectric, Fluorescence and Thermal analysis have been reported for various applications.

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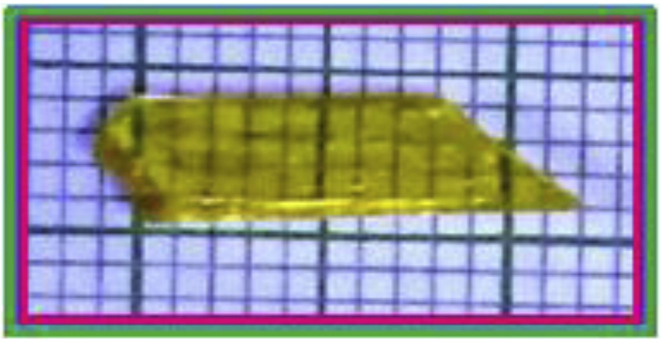

Fig. 1. The photograph of grown crystal BCDH.

2. Material synthesis and growth of BCDH

BCDH crystal was grown by taking commercially available sample of Brucine (Sd. Fine: 99%, AR grade) and Copper (II) chloride (Merck - 99%) in the equimolar ratio (1:1) by slow evaporation method at ambient temperature. Initially, the calculated amount of Brucine and the Cupper (II) chloride are completely dissolved separately by using the solvents double distilled water-ethanol (1:1) and water respectively. By mixing both the solutions together, a green precipitate was obtained. As prepared salt was dried and dissolved by utilizing ethanol-water as a solvent. Then, the solution was stirred well about 4 h at ambient temperature and the saturated solution was filtered with the help of Wattman (grade no.1) filter paper in a clean beaker. The pH value of prepared solution is found to be 5 it was directly measured using pH indicator paper. The beaker containing the solutions were covered with perforated polythene sheets and kept in undisturbed place for the purpose of solvent evaporation. After 40–45 days, high transparent crystal was harvested from the mother solution and it is displayed in Fig. 1.

2.1. Experimental details

The single crystal XRD analysis of BCDH with dimension $0.3 \times 0.2 \times 0.25\ mm^3$ was evaluated using Bruker AXS Kappa APEX III CMOS Diffractometer furnished with graphite monochromated Cu Kα radiation (λ = 1.5418 Å) at room temperature. CHN analysis of the BCDH was carried out utilizing Elementar Vario EL III-(Germany) instrument. Morphological analysis of the BCDH crystal was identified by the single crystal XRD study. The UV–Visible spectrum was recorded by using Shimadzu UV 1600 spectrometer in the ranges between 200 and 800 nm. The both FT-IR and FT-Raman spectrum spectral investigation was done using BRUKER IFS 66 spectrometer and FT-Raman Bruker RFS 27: S Raman module at room temperature respectively. The piezoelectric charge coefficient (d_{33}) of the BCDH compound was measured out by PM-300 piezometer at ambient temperature with the calibrated force of 0.25 N. A Q-switched High Energy Nd: YAG Laser (QUANTA RAY Model LAB–170-10) beam of wavelength 1064 nm and 6 ns pulse width with the periodicity frequency of 10 Hz and energy 0.70 mJ/pulse was used for NLO and LDT measurements. The Fluorescence spectral study and lifetime quantification were carried out with the help of FLUOROCUBE (JOBIN-VYON M/S) spectro-fluorometer with the wavelength of 317 nm. The thermal analysis of BCDH was carried out in the temperature range from 25 to 1400 °C in an alumina crucible in the Nitrogen environment using NETZSCH STA 449F3 at a heating rate of 20 °C/min.

3. Results and discussion

3.1. Crystal structure analysis

The unit cell parameters were quantified from the reflections of 36 frames measured in 3-various crystallographic zones by the approach of difference vectors. After the careful examination of quality and unit cell parameters of the crystal were set for data collection using ω-φ scan modes. The process of data collection, reduction and absorption correction were done by the programs APEX2, SAINT-plus and SADABS [13] respectively. The structure of the compound BCDH was resolved by direct method procedure using SHELXS-2016 program and refined by Full-matrix least square procedure on F^2 using SHELXL-2016 program [14]. All the non H-atoms were subjected to anisotropic refinement whereas the H-atoms were refined isotropically.

The structure was initially refined to a high R index of 17.8% and the difference Fourier map show relatively larger peaks ($\Delta\rho_{max}$ = 1.32 $eÅ^{-3}$) and showing poor convergence. A preliminary check with TwinRotMat routine of PLATON [15] shows that, the BCDH crystal possess the 2-fold twinning about (0 0 1) with the twin matrix of (−1 0 0.003/0–1 0/0.5 0 1). After applying the twin correction [16] the structure refined to an improved R index of 8.78% and with a considerably flatter difference Fourier map ($\Delta\rho_{max}$ = 0.53 $eÅ^{-3}$) and BASF of 0.432.

In the structure, one of the Cl anion and water molecules are experience a positional disorder. The positions of all the H-atoms were initially identified from the difference electron density map and were treated accordingly. The H-atoms bounded with C-atoms were treated as riding atoms with distances of d(C-H) = 0.96 (for CH_3) with Uiso (H) = −1.5Uequ(C), d(C-H) = 0.97 Å (for CH_2) with Uiso (H) = −1.2Uequ(C), and d(C-H) = 0.93 Å (for aromatic CH) with Uiso (H) = −1.2Uequ(C). The H-atom associated to protonated N-atoms were identified from the difference electron density peak and were geometrically optimized. The O-atoms associated with all the H-atoms were identified from the difference electron density map and were restrained to a distance of d (O-H) = 0.90 Å with s.u. of 0.01 Å. The crystal structure refinement details are listed in Table 1. The molecular graphics was done using the software ORTEP [17] and Mercury [18] software's. The structure (i.e. ORTEP) of the compound BCDH is shown in Fig. 2. The asymmetric unit comprises two brucinium cations, two chlorine anions and two water molecules due to the conformation flexibility resulting a set of eight molecules in the unit cell. One of the Cl anion and water molecules is disordered at two positions (0.55:0.45)

Table 1
Crystal data and structure refinement for BCDH.

Identification code	BCDH
CCDC No.	1823610
Empirical formula	$C_{46}\ H_{61}\ Cl_2\ N_4\ O_{12}$
Molecular weight	932.88
Temperature	291(2) K
Wavelength	1.54178 Å
Crystal system, space group	Monoclinic, C2
Unit cell dimensions	a = 14.1700(5) Å; α = γ = 90° b = 12.3340(4) Å; β = 97.570(2) ° c = 26.3458(9) Å
Volume	4564.4(3) $Å^3$
Z, Calculated density	4, 1.358 $Mg\ m^{-3}$
Absorption coefficient	1.840 mm^{-1}
F (000)	1980
Crystal size	$0.3 \times 0.2 \times 0.25\ mm^3$
Theta range for data collection	4.771–67.297°
Limiting indices	$-16 \le h \le 16$, $-14 \le k \le 14$, $-31 \le l \le 31$
Reflections collected/unique	26449/26449 [R(int) = 0.0000 ?]
Completeness to theta = 67.297	99.00%
Refinement method	Full-matrix least-squares on F^2
Data/restraints/parameters	4249/16/632
Goodness-of-fit on F^2	1.079
Final R indices [I > 2sigma(I)]	R1 = 0.0877, w*R*2 = 0.2094
R indices (all data)	R1 = 0.1279, w*R*2 = 0.2408
Absolute structure parameter	0.076(13)
Largest diff. peak and hole	0.528 and −0.329 e. A^{-3}

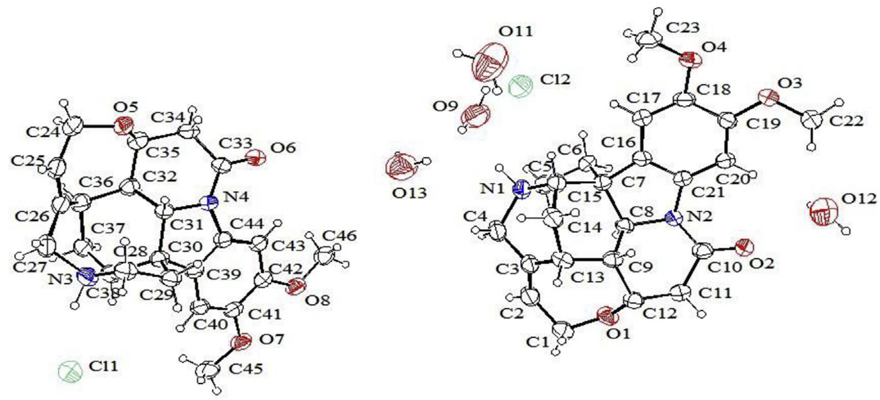

Fig. 2. Molecular structure (ORTEP) of the compound BCDH drawn at 40% probability level.

as discussed in the experimental collection section.

In the crystal structure the brucinium cations and the Cl^- anions along with the lattice water molecules form interesting supramolecular assembly through potential hydrogen bonds such as N-H⋯O, O-H⋯O, N-H⋯Cl and C-H⋯O. Of the protonated nitrogen atoms of both the molecules, one molecule forms hydrogen bond with water oxygen atom (N1-H1⋯O9) whereas the other molecule generates hydrogen bond with the Cl^- anion (N3-H3⋯Cl). These intermolecular interactions are of great importance in exhibiting NLO and other physiochemical properties in spite of the non-centric or non-centrosymmertic nature of the crystal. The adjacent brucinium cations are linked through C-H⋯O hydrogen bonds forming a zig-zag 1D-supramolecular chain extending along the (0 1 0) crystallographic direction as shown in Fig. 3. Both the molecules of the asymmetric unit form independent (0 1 0) chain separated by a distance of half the length of the c-axis i.e. at (0 0 0) and (0 0 ½). Interestingly the adjacent one dimensional (0 0 1) chains stacked along the c-axis are bridged by the lattice water molecules and chlorine anion through a variety of electrostatic interactions such as N-H⋯O, N-H⋯Cl and water O-H⋯O hydrogen bonds. This forms a 2D-supramolecular sheet extending parallel to the (1 0 0) plane and is shown in Fig. 4. The Cl anions are positioned in the void between the 1D-supramolecular chain which acts as a potential H-bond acceptor for the C and N acceptor of the brucinium molecule. Molecular packing of the crystal is stabilized by weak O⋯Cl and O⋯O through a 3D-network as depicted in Fig. 5. The H-bonding geometry of the BCDH crystal was listed in Table 2.

The space group was deduced to be non-centrosymmetric C2 of monoclinic crystal system with lattice parameters are a = 14.17 Å, b = 12.33 Å, c = 26.34 Å, α = γ = 90°, β = 97.57° and its unit cell volume is V = 4564.4(3) $Å^3$. The seven membered ring (C1-C2-C3-C13-C9-C12-O1) of BCDH is in chair conformation. Similarly, two six membered rings (C16-C17-C18-C19-C20-C21) and (C13-C14-C15-N1-C4-C3) of BCDH were planar and boat conformation respectively.

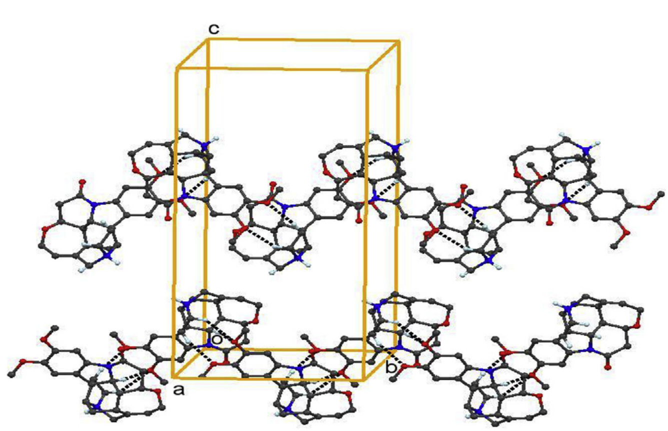

Fig. 3. Part of the crystal structure of BCDH showing the formation of one-dimensionalsupramolecular
chain along (0 1 0) direction through C-H⋯O hydrogen bonds.

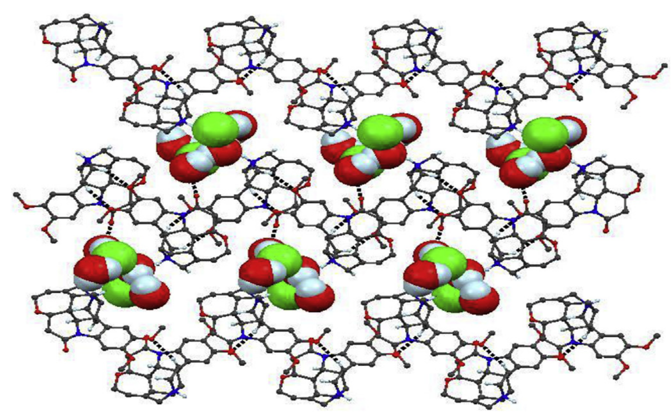

Fig. 4. Part of the crystal structure of BCDH showing the formation of two-dimensional supra molecular sheet extending parallel to the (1 0 0) plane. The Cl anions and the lattice watermolecules occupying the space is represented as space filled form.

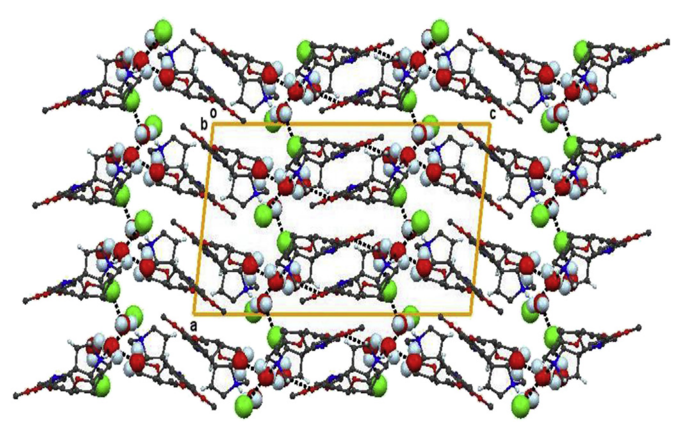

Fig. 5. Molecular packing of the crystal viewed along b-axis.

3.2. Chemical CHN analysis

The concentration of carbon (C), hydrogen (H) and nitrogen (N) was calculated by the standard the CHN analyzer. The presence of atomic weight percentage values of C, H and N in the BCDH crystal was measured and compared with theoretical values. The result concludes

Table 2
Hydrogen bond geometry of BCDH.

D-H ... A	d(D-H)	d(H ... A)	d(D ... A	< (DHA)
C4-H4B···O11[i]	0.97	2.63	3.35(3)	130.8
C5-H5B···O3[ii]	0.97	2.55	3.387(14)	144.7
C6-H6B···O4[ii]	0.97	2.52	3.430(13)	157.1
C15-H15···Cl2	0.98	2.81	3.762(13)	165
C29-H29B···O7 [iii]	0.97	2.51	3.429(14)	158.6
N1-H1···O9	0.98	1.85	2.781(19)	158.4
N3-H3···Cl1	0.98	2.09	3.030(11)	160.7
O13-H13C···Cl1 [iii]	0.900(4)	2.94(19)	3.76(3)	152(31)
O10-H10B···Cl1 [iii]	0.900(3)	2.40(13)	3.143(19)	140(16)
O9-H9B···O12[iv]	0.90(3)	2.05(11)	2.93(3) 163(33)	
O9-H9A ... O3	0.90(3)	2.2(2)	2.91(4)	139(26)
O12-H12A ... Cl2 [ii]	0.900(4)	2.21(11)	3.02(3)	149(19)
O12-H12B···O9[ii]	0.900(3)	2.19(10)	2.93(3)	138(13)

Symmetry code:
(i) x+1/2,y-1/2,z (ii) -x+1/2,y-1/2,-z+1 (iii) -x+1/2,y+1/2,-z(iv) -x+1/2,y+1/2,-z+1.

that, BCDH crystal possesses C = 36.50% (36.81%), H = 49.03% (48.88%) and N = 3.17% (3.20%). The value in bracket represents the theoretical composition and which is confirmed that well agreement with the experimental values.

3.3. Morphological study

Growth mechanism and morphological behavior of the grown crystals are very important in consummating and manipulating many of its physicochemical properties. The morphological behavior of the grown crystal depends on various parameters such as nature of the solvent, pH of the solution, molecule interactions between solute and solvent, super-saturation state of the solution and adsorption of the planes, etc. And the development faces of crystal depend on growth velocity and presence of impurities in the solution. Morphological study of the BCDH reveals that it has seven developed faces with major faces (0 0 1), (0 0–1), (1 1 -1) and (-1 -1 1). Compare to the growth directions along with a-axis and b-axis the growth rate along c-axis was high, which is clearly shown in Fig. 6.

3.4. Linear optical study

The optical properties of materials furnishes the information about the cut-off wavelength, band gap energy (E_g), absorption co-efficient, reflectance, excitation co-efficient, refractive index, etc. Also, it is closely associates with the impurity levels, lattice vibrations, atomic structure, electronic band structure and electrical properties. The lesser cut-off wavelength values and higher optical transmittance range of the materials are possess the higher SHG efficiency. The UV transmittance spectrum of the BCDH crystal is given in Fig. 7a. It is noticed that, the grown crystal has wide transmittance (and low absorbance) in the entire Visible range. The 'lower cut-off' wavelength of BCDH crystal was 317 nm which is identified from Fig. 7a. This lesser cut-off wavelength was obtained due to the promotion of an electron from a 'non-bonding' (lone-pair) n orbital to an 'antibonding' π orbital and this transition

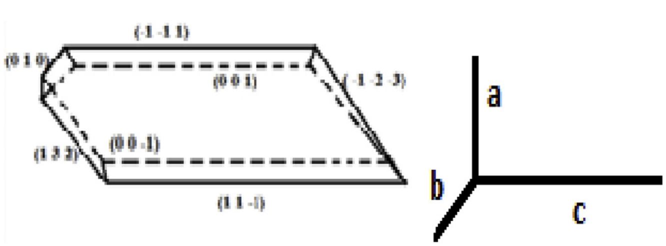

Fig. 6. Morphological diagram of title crystal.

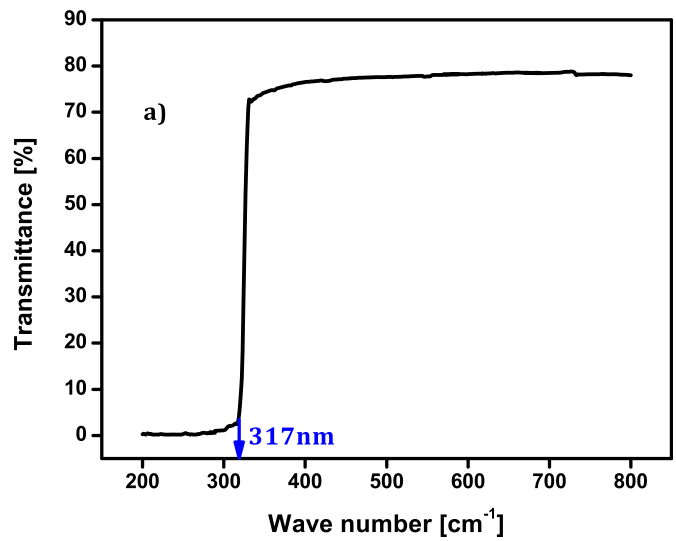

Fig. 7a. Transmittance spectrum of BCDH.

simply designated as $\pi*(n \rightarrow \pi*)$ and also straight line portion beyond the 317 nm shows that the no absorption was remarked in the entire visible range with prominent transmittance (≈78%), hence BCDH has good crystal quality finds suitability in optical communication. The absorption coefficient (α) was calculated directly using the relation:

$$\alpha = \frac{2.3026}{t} \log_{10}\left(\frac{1}{T}\right) \tag{1}$$

where, T and t are the transmittance and thickness of the crystal respectively. The band gap (E_g) energy of the material can be measured from cut-off wavelength by using the relation:

$$E_g = \frac{hC}{\lambda_c} \tag{2}$$

where, E_g is optical band gap, h is Plank's constant (6.626×10^{-34} J), C is velocity of light and λ_c is cut-off wavelength. On substituting these values in eq. (2) then the value of E_g was found to be 3.91 eV. The quantity absorption co-efficient (α) can be evaluated by the Tauc's relation:

$$\alpha h\nu = \beta(h\nu - Eg)^m \tag{3}$$

The value of E_g and m corresponds to the energy and the nature of the particular optical transition with absorption co-efficient α. For optical transitions such as direct allowed, direct forbidden, indirect allowed and indirect forbidden transitions the theoretical value of 'm' has 1/2, 3/2, 2 and 3 respectively [19]. Band gap energy plays an important role in the determination of electrical conductivity of solids. Fig. 7b shows the Tauc's plot between product of absorption coefficient and the incident photon energy $(\alpha h\nu)^{1/2}$ with the photon energy ($h\nu$) at room temperature depicts a linear behavior, which can be regarded as an evident that the direct transition. Hence, assuming a direct transition between valence and conduction bands, the optical energy band gap (E_g) is estimated by extrapolation of the linear portion of the curve to the point $(\alpha h\nu)^{1/2} = 0$. The E_g of the crystal is found to be 3.80 eV. It is fairly agree with the value calculated by using eqn. (2) i.e 3.91 eV. The large value E_g showed that the defect compactness in the BCDH crystal is very low [20] and it is very desirable prospect for optoelectronic applications [21].

3.5. Vibrational analysis

FTIR and FT-Raman spectroscopic studies are utilized to classify the functional groups uniquely and to determine the molecular configuration of the synthesized compound. These vibrational methods can

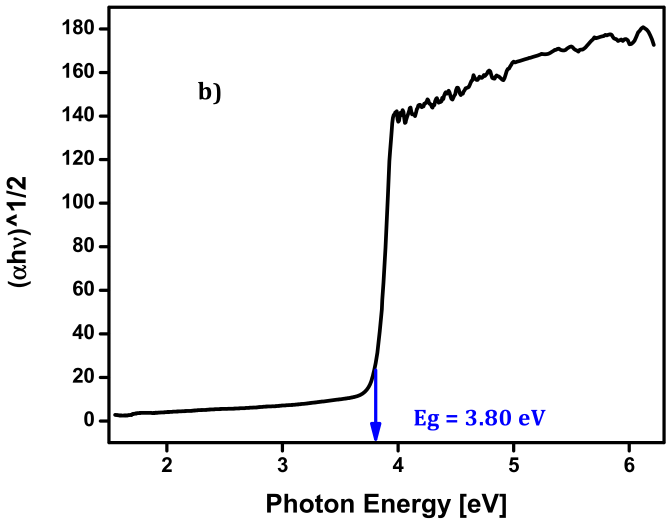

Fig. 7b. Plot of $(\alpha h\nu)^{1/2}$ vs photon energy.

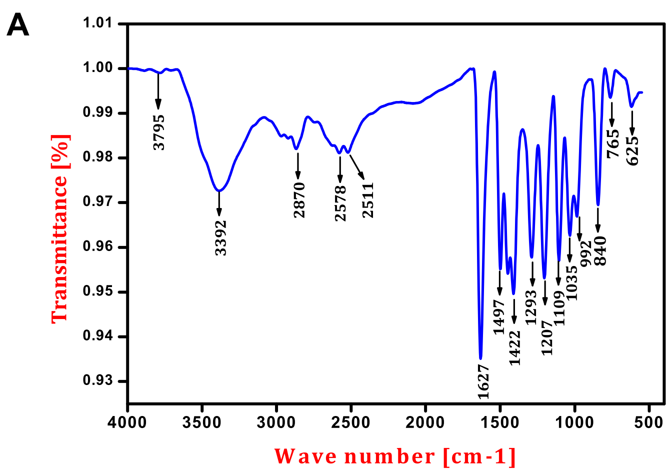

Fig. 8a. FTIR-spectrum of BCDH.

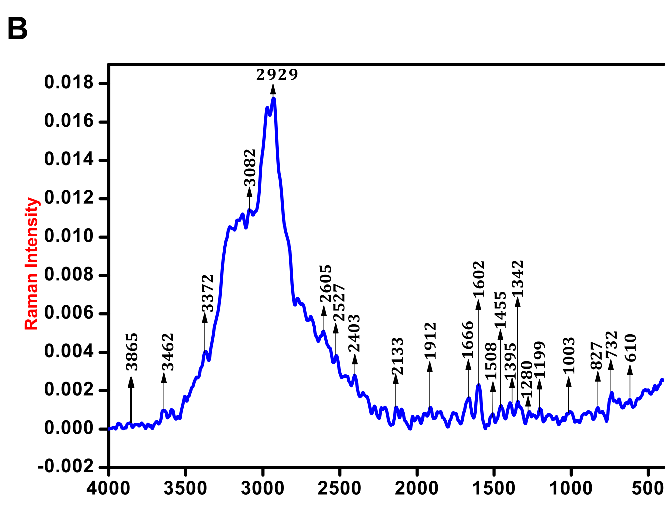

Fig. 8b. FT-Raman spectrum of BCDH.

provide valuable information about the force field and can help to understanding the physiochemical properties in solid states. The observed FT-IR and FT-Raman spectrum of the powdered sample BCDH crystal at room temperature are dissipated in Fig. 8a and b respectively.

3.5.1. O-H Vibration

The position of OH band is mainly depends on the vigor of presence of hydrogen bonds in the compound. The frequencies of bending and stretching vibrations and force constant of the both acceptor and donor groups are altered by the H-bonding. The OH vibrations are usually observed in the spectral range 3600 to 3400 cm^{-1} [22]. The weak band observed at 3795 cm^{-1} in FT-IR is assigned to O-H symmetric stretching vibration. In the FT-Raman spectrum this matching part of vibration was occurred at 3865 cm^{-1}.

3.5.2. N-H Vibration

Commonly, in the vibrational spectra the frequency range 3500 to 3000 cm^{-1} the N-H vibrations are occurs [23]. In this work, a strong band at 3392 cm^{-1} in the FT-IR was identified the N-H symmetric stretching vibration of the BCDH crystal and this corresponding peak was observed in FT-Raman at 3372 cm^{-1}.

3.5.3. C-H Vibration

The title compound contains a methyl group attached to the aromatic ring. In aromatic molecules, the CH_3 symmetric stretching and CH_3 asymmetric stretching vibrations are arises in the frequency range 2870 to 2860 cm^{-1} and 3000 to 2905 cm^{-1} respectively [24,25]. In the present study, the peak attributed at 2870 cm^{-1} in FT-IR and 2929 cm^{-1} in FT-Raman spectrum was due to existence of the symmetric stretching vibration of CH_3. In FT-IR spectrum, the characteristic peaks at 2578 cm^{-1} and 2511 cm^{-1} are assigned C-H symmetric stretching vibrations of BCDH and its corresponding peaks were observed in FT-Raman at 2605 cm^{-1} and 2527 cm^{-1} respectively. The structure of the hetro aromatic compounds and also most of its derivatives are similar to that of benzene. Mostly, C-H out-of-plane and in-plane bending vibrations are observed in the frequency range 950 to 600 cm^{-1} and 1300 to 1000 cm^{-1} respectively [24,26]. For different vibrational frequencies such as 1293 cm^{-1} and 1207 cm^{-1} in the FT-IR were due to the existence of C-H in-plane bending vibrations. And also, its corresponding counter parts were arises at the frequencies 1280 cm^{-1} and 1199 cm^{-1} respectively. Similarly, the characteristic band observed at 840 cm^{-1} and 765 cm^{-1} in the FT-IR are assigned to C-H out-plane bending vibrations. The FT-Raman counterpart arising from these vibrations were occurred at 827 cm^{-1} and 732 cm^{-1} respectively.

3.5.4. C-C and C=C vibrations

In general, the vibration between two carbon atoms are occurs in the spectral range 1650 to 1000 cm^{-1}. Especially, the C=C and C-C stretching vibrations of the aromatic rings are take place in the frequency range 1650 to 1430 cm^{-1} [27] and 1600 to 1000 cm^{-1} [28] respectively. The C=C stretching vibrations are assigned to the modes 1627 and 1497 cm^{-1} in the FT-IR and 1602 and 1508 cm^{-1} in FT-Raman spectra. In the same way, a peak attributed at 1422 cm^{-1} in FT-IR was due to C-C symmetric stretching vibration of BCDH compound and its corresponding peak was obtained in FT-Raman at 1455 cm^{-1}.

3.5.5. C-N Vibration

The C-N vibrations are difficult task, because mixing vibrations are most probable in this spectral region. In general, C-N stretching mode of vibrations are appear in the spectral range of 1150 to 950 cm^{-1} [29]. In the present case, the band of frequency at 992 cm^{-1} in FT-IR and 1003 cm^{-1} in Raman are corresponds to the C-N stretching vibration.

3.5.6. C-O Vibration

The FT-IR band at 625 cm^{-1} and its Raman counterpart at 610 cm^{-1} were attributed to the C-O asymmetric stretching vibrations [30]. The important FT-IR and FT-Raman peak values (i.e. wave numbers in cm^{-1}) and its tentative assignments of the title BCDH compound were listed in Table 3.

Table 3
The various functional group vibrations and their tentative assignments.

Assignments	Wave number (cm^{-1})	
	FT-IR	FT-Raman
O-H symmetric stretching	3795	3865
N-H symmetric stretching	3392	3372
CH_3 symmetric stretching	2870	2929
C-H symmetric stretching	2578, 2511	2605, 2527
C=C stretching	1627, 1497	1602, 1508
C-C symmetric stretching	1422	1455
CH in-plane bending	1293, 1207	1280, 1199
C-N stretching	992	1003
C-H out-plane bending	840,765	827,732
C=O out-plane bending	625	610

3.6. Piezoelectric measurement

Piezoelectricity is the accumulation of electric charges by applying mechanical stress and it is closely related to the dipole moment and crystal symmetry of solids. Piezoelectric materials are widely used in many ways such as inkjet printers, discovery and production of sonar waves, single-axis and dual-axis tilt sensing, acceleromyography (detection of muscle movements), piezosurgery (cut a aimed tissue with slight damage to adjacent tissues) [31,32], etc. Piezoelectric properties are largely affected by the dislocations [33]. The Piezoelectric charge co-efficient (d_{33}) of the BCDH was measured directly from cathode ray oscilloscope which gives 4 p C/N. Hence, the BCDH crystal would be capable candidate for sensing microstructures on electronic microchips, electric voltage sources, piezo-sensors and transducers applications [34,35] etc.

3.7. Second harmonic generation (SHG) studies

The nonlinear optical property and frequency doubling efficiency of the BBDH crystal was confirmed with the help of Kurtz and Perry powder method. The output of SHG are depends on the particle size, charge separation/transfer, intermolecular interactions and its structural attributes [36–38]. In general, more hydrogen bonded materials are possess the efficient piezoelectric and SHG properties [39,40] and hence due to presence of more hydrogen bonds in the title crystal we can expect higher piezoelectric and SHG properties. Most of the non-centrosymmetric crystals would exhibit the SHG whereas third harmonic generation (THG) is not a prime factor [37,41,42]. The powdered sample of the BCDH crystal was tightly filled in between the two glass plates. An elemental laser beam of wavelength 1064 nm from a Nd: YAG laser was made to incident on the powder sample cell and SHG was confirmed by emanation of green light (λ = 532 nm). In this process, the frequency of the output beam was doubled. The quantified SHG efficiency of BCDH compound was found to be 1.70 times greater than that of the standard KDP. Hence, the grown crystal can be used as frequency doublers and in addition to that the potential candidates for NLO applications and device geometries [37].

3.8. LDT studies

LDT is an effective method for measuring the amount of electromagnetic radiation an optical component that can resist. A various parameters such as location of the beam, incident wavelength, input energy pulse, pulse duration, beam size, transverse and longitudinal mode structure, experiment geometry, and repetition rate, etc. which is influences the LDT value. Normally, dislocations free materials are having large LDT value. The performance of the crystal primarily depends on the surface superiority and the surface damage of the crystal was quantified utilizing the following equation:

$$\text{Energy density } (E_d) = \frac{\mathrm{E}}{\tau A} \tag{4}$$

where 'E' is input energy (mJ) and 'A' area of the spot (mm^2). The calculated LDT value of the BCDH crystal was 3.51 GW/cm^2. It is noticed that the determined LDT values are 10.96, 17.55 and 2.34 times higher than that of standard materials such as KTP (0.32 GW/cm^2), KDP (0.2 GW/cm^2) and urea (1.5 GW/cm^2) [43] respectively. Also, some of the brucinium based compounds particularly B5ST [44], BHF [45], BHM [46], 2,3D10-OHOD [47], B2C6ND [48] and DBSH [9] are possess the LDT values 13.64, 13.64, 8.6, 6.8, 2.8 and 2.3 GW/cm^2 respectively. The higher LDT value of BCDH shows that, it is a useful candidate for fabrication of high-power laser application devices [10].

3.9. Fluorescence studies

The process of fluorescence can be expected from the aromatic compounds or molecules containing double bounds with higher order resonance stability [49]. It finds wide applications in various fields for analyzing organic compounds such as medical, chemical and biochemical research. Likewise, it is very useful in the field of optoelectronics, particularly in light emitting diodes and solid state lasers [50]. The wavelength (i.e energy) of the emitted light is depended on the optical band gap. The complete energy equilibrium for the process of fluorescence could be written of the following equation as:

$$E_{fluor} = E_{abs} - E_{vib} - E_{solv.relax} \tag{5}$$

where, E_{fluor} , E_{abs} and E_{vib} are energy of the emitted light, energy of the light absorbed and energy lost by the molecule respectively and $E_{solv.relax}$ is the solvent cage of the molecule. Fluorescence spectroscopy is one of the branches of electromagnetic spectroscopy that examines fluorescence from a sample and it is primarily related with electronic and vibrational states and this is useful tool for measure the pH inside of the cells and cellular compartments. Fluorescence spectra of the materials are affected by many factors namely solvent polarity and viscosity, rate of solvent relaxation, internal charge transfer, proton transfer, excited state reaction and probe-probe interaction, etc.

Fig. 9 depicts that the fluorescence profile of title compound at excitation wavelength 317 nm (i.e 3.91 eV) was studied to outlook over the emission spectrum in regard with that particular excited state of the system and decaying components [51]. The emission peak was occurring at 412 nm (i.e. 3.01 eV), which lies in the UV–visible range accordance with the electromagnetic spectrum. The other less intense peak assigned at 433 nm (i.e. 2.86 eV) near to emission peak may due to presence of defect inside the grown crystal. The sharp and high intense

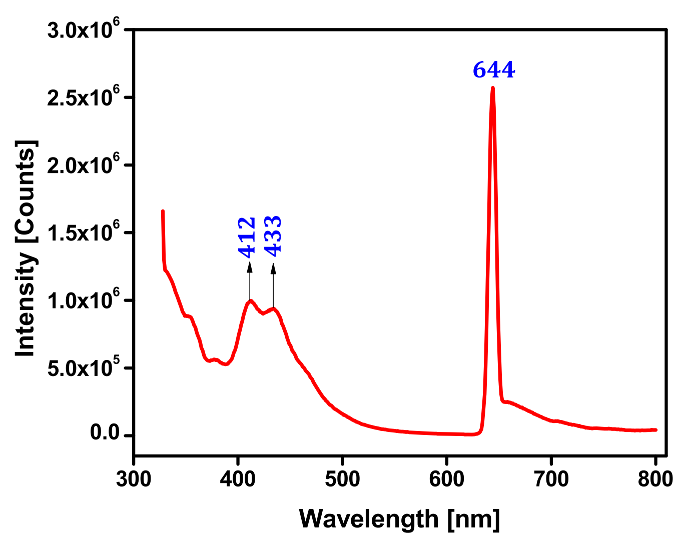

Fig. 9. Emission spectrum of BCDH.

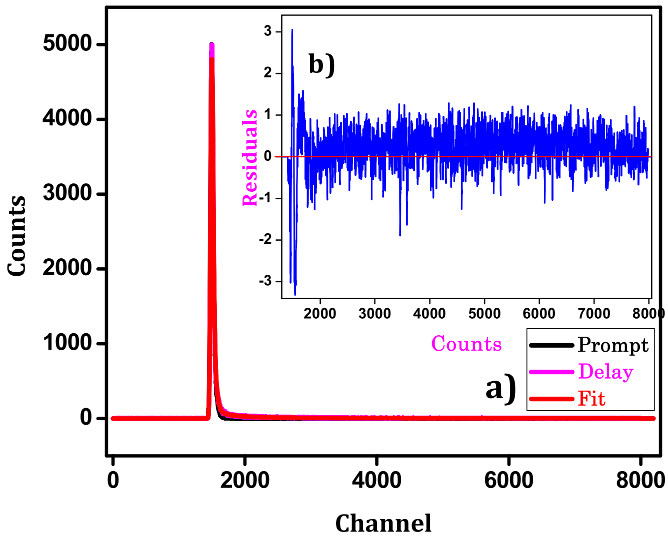

Fig. 10. (a) Fluorescence lifetime spectrum (b) [Inset] Residual fit of BCDH.

peak appear at 644 nm (i.e 1.92 eV) is attributed to red fluorescence emission. Hence, the BCDH crystal may useful utensil for OLED applications [52].

3.9.1. Life time (τ) measurements

The lifetime defined as mean time of the molecule spends in the higher state before returns to the lower state and it is independent of probe concentration. The lifetime profile was calculated by Time-Correlated Single Photon Counting (TCSPC) method. Fig. 10a unveils that, the three decay time spectrum of the BCDH crystal. The analysis of decay time measurement was fitted with the two exponential decay components:

$$F(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + A_3 e^{-t/\tau_3} \quad (6)$$

where, A_1, A_2 and A_3 denote amplitudes and τ_1, τ_2 and τ_3 are lifetimes of prompt and delayed emissions. An extent to which the most beneficial fitting could be performed to the actual decay curve is well concluded from the residual fit as shown in Fig. 10b. The perfectness of the curve fit could be measured from the residuals, and the value of the reduced nonzero second-order molecular polarizabilty (χ^2) ratio. The deliver examination presented the shortest decay component of lifetime and amplitude is detailed in Table 4. Finally, from the various optical analysis (linear and non linear optical study, LDT, Fluorescence) suggested that the BCDH crystal might be appropriate candidates for numerous applications such as laser Q-switching, laser mode-locking, optical mouse, optical bistability, night vision devices, two-wave mixing, holographic recording [53], attenuator, optical amplifier, and optical diode applications [54].

3.9.2. Thermal analysis

Thermal and physicochemical behaviours of non linear optical BCDH crystal were analyzed by TGA (Thermal gravimetric Analysis), DTA (Differential thermal analysis) and DSC (Differential Scanning Calorimetry) studies. For this analysis, 2.797 mg of BCDH compound was used. The TGA and DSC analysis of the title compound is shown in Fig. 11. It reveals that the first decomposition occur at 102.31 °C in TGA

Table 4
Lifetime analysis of BCDH crystal by Fluorescence study.

Sample code	Analysis	Lifetime (ns)			Amplitude			χ^2
BCDH	Three Exponential	τ_1 0.83	τ_2 1.67	τ_3 3.35	A_1 3.84	A_2 6.08	A_3 90.08	1.14

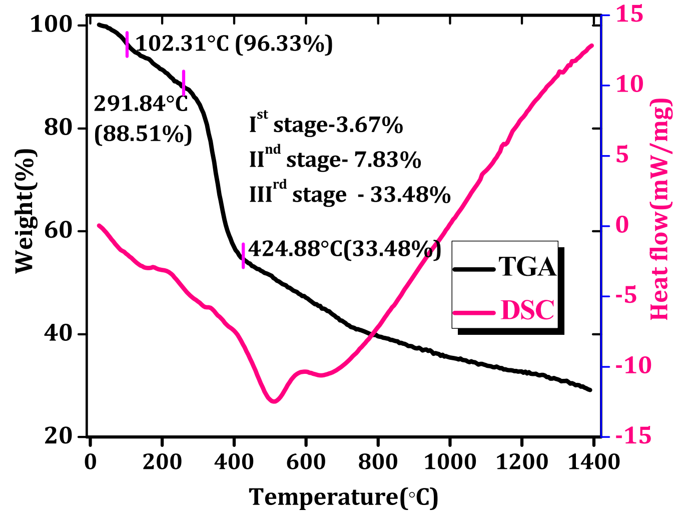

Fig. 11. TGA and DSC curves of BCDH compound.

is due to the removal of water molecules and also chlorine ions because of its hesitant interaction with brucine and which is point out that the thermal stability of the compound. The second stage of decomposition was observed at 291.84 °C and this is due to the broken of C–O and C–C bonds in brucine [47]. Similarly in the third stage the remaining brucine ion was decomposed at 424.88 °C. The percentages of weight losses for various steps are depicted at the center of Fig. 11. This is in good compliance with that pragmatic in DSC curve.

The DTA analysis curve of the title compound is shown in Fig. 12. The DTA endothermic peak was observed at 104.28 °C corresponds to the melting point of the BCDH compound which is confirms the stability of the compound. The DTA peak is concur with TGA curve. The exothermic sharp peak of BCDH crystal was confirms that existence of purity and good degree of crystallinity of the sample [55].

4. Conclusion

The BCDH crystals were grown by slow evaporation technique and it belongs to non-centro symmetric space group C2 of monoclinic crystal system which is confirmed by single crystal XRD studies. The presences of various chemical elements were confirmed by the CHN analysis. Morphological analysis of the BCDH reveals that it possess seven developed faces with four major faces (0 0 1), (0 0–1), (1 1 -1)

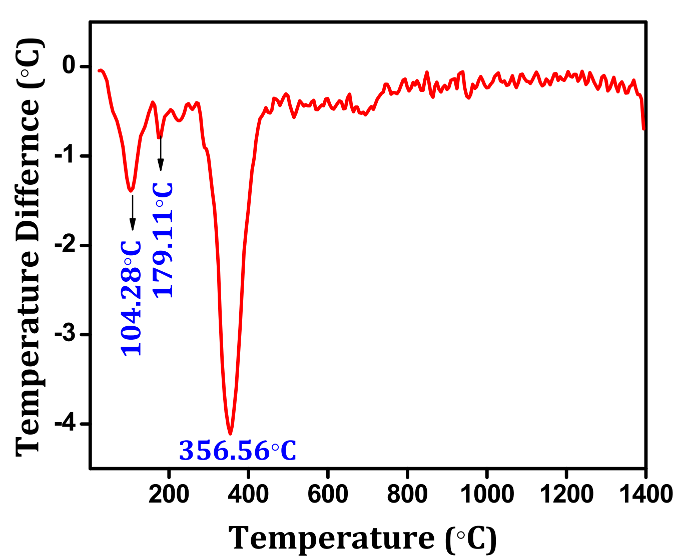

Fig. 12. DTA curve of BCDH crystal.

and (-1 -1 1). The UV–Visible spectrum showed that, it has 78% transparency in the visible region. The presence of variance functional groups of title compound were confirmed by FTIR and FT-Raman analysis. Piezoelectric measurement suggested that the BCDH crystal is promising candidates for transducers and piezo-sensors applications. The SHG output of the BCDH crystal was measured by Kurtz and Perry powder method. The LDT value of the BCDH crystal was calculated and also compared with some other standard materials. Fluorescence spectral study revealed the electron excitation in the grown BCDH crystal. The thermal performances of the BCDH compound were scrutinized by TGA, DTA and DSC analysis.

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